PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

MOROGORO MUNICIPAL COUNCIL

FORM THREE TERMINAL JOINT EXAMINATION

CODE: 032 CHEMISTRY

TIME: 3HRS MAY, 2025

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INSTRUCTIONS:

1. This paper consists of sections **A, B and C** with a total of **eleven (11)** questions
2. Answer all questions form section **A** and **B** and only **two (2)** questions from section **C**
3. Section **A** carries sixteen (16) marks, section **B** carries **fifty-four (54)** marks and section **C** carries **thirty (30)** marks.
4. All writings must be in blue or black ink with exception of diagrams which must be done in pencil
5. Cellular phones and any unauthorized materials are not allowed in the examination room
6. Remember to write your Name and stream on every page of your answer sheet(s)
7. Whenever necessary use the following constants
   1 faraday = 96500C
   1 litre =1dm^3 = 1000cm^3
   Avogadro's constant = 6.022×10^{22} $particles/mole$
   G.M.V. at S.T.P. = 22.4 dm^3
   Standard pressure = 760mmHg
   Standard temperature = 273K
   Atomic masses
   $H = 1, C = 12, O = 16, Na = 23, Al = 27, S = 32, Cu = 64, Ag = 108$ Pb =207

SECTION A (15 Marks)
Answer all questions in this section

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the booklet provided.

(i) The pair of elements which is most likely to form a covalent bond when reacted together is

A. Sodium and iodine
B. Magnesium and oxygen
C. Aluminium and oxygen
D. Carbon and chlorine
E. Calcium and carbon

(ii) Sea water contains various salts. Which salt is present in the greatest proportions?

A. Magnesium chloride
B. Calcium sulphate
C. Potassium chloride
D. Sodium chloride
E. Magnesium sulphate

(iii) The process of distillation involves

A. Filtration and decantation
B. Evaporation and condensation
C. Evaporation and filtration
D. Condensation and decantation
E. None of the above

(iv) Elements are classified as metals and non-metals. Which of the following electronic configuration is of metal?

A. 2:8:8:1 and 2:8:8:7
B. 2:8:3 and 2:8:6
C. 2:8:8:1 and 2:8:3
D. 2:8:6 and 2:8:8:7
E. 2:5 and 2:4

(v) Some antacids contain magnesium hydroxide, which reacts with stomach acid (aqueous hydrochloric acid) as shown by the equation

$Mg(OH)_{2(aq)} + 2HCl_{(aq)} \longrightarrow MgCl_{2(aq)} + 2H_2O$

This is an example of

A. Thermal decomposition
B. Double decomposition
C. Neutralization
D. Precipitation
E. Acid – base reaction

(vi) The elements Helium, Neon, Argon and Krypton are known as

A. Alkali metals
B. Halogens
C. Inert gases
D. Non metals
E. Transition elements

(vii) The magnesium salt responsible for permanent hardness of water is the

A. Hydrogen carbonate
B. Sulphate

C. Nitrate
D. Carbonate
E. Cyanide

(viii) The similarity between rusting of iron and burning of magnesium in air is that both
A. Use carbon dioxide
B. Are reduction processes
C. Do not need water
D. Use up oxygen
E. Are redox compound

(ix) Iron, when exposed to moist air rusts. Which one of the following metals is used to coat iron so as to prevent it from rusting so that it is then known galvanized iron?
A. Zinc
B. Tin
C. Copper
D. Silver
E. Lead

(x) Which set contains three substances which will lose weight when heated because of dehydration?
A. Pb, Fe and Cu
B. CuO, NaO and ZnO
C. Pb (NO_3), $KClO_3$ and $CuCO_3$
D. C_2H_5OH, $Na_2CO_3.10H_2O$ and $FeSO_4.7H_2O$
E. $NaHCO_3$, CuO and $ZnCO_3$

2. Match the items in List A with the responses in List B by writing the letter of the correct response in the answer sheets provided.

LIST A		LIST B
(i)	Used to pour a solution into the burette without spilling it	A. Pipette B. Pipette filter C. White tile D. Retort stand and clamp E. Conical flask F. Burette G. Funnel H. Dropper
(ii)	Used to hold the burette in an upright position while performing volumetric analysis experiments	
(iii)	It is used to be placed at the base of the apparatus holds the burette so that to have clear vision	
(iv)	It is used to deliver acid solution during titration process	
(v)	Used to measure and transfer definite volume of solution during titration process	
(vi)	It has a neck which easily held while swirling the contents inside	

SECTION B (54 Marks)
Attempt all questions from this section

3. (a) Mr. Big was provided with list of four atoms whose mass and atomic numbers have be indicated as $^{37}_{17}X$, $^{35}_{17}Y$, $^{32}_{16}W$ and $^{32}_{18}W$. Which of the atom(s) above represent?
(i) Isotopes? Explain why?
(ii) What is the number of protons in X?

(iii) What is the number of neutrons in Y?

(b) There were two beakers, **A** and **B** in the laboratory which contained water. Then after experimental analysis it was found that beaker **A** had calcium hydrogen carbonate and beaker **B** had sulphate of magnesium. As an expert of chemistry suggest,

(i) The types of water found in the laboratory

(ii) Using equation, show how the hardness found in the water can be removed in the two beakers (one point)

4. (a) During a chemistry session, Sir Constantino told his students that, "When an aqueous solution of silver nitrate is mixed with aqueous solution of calcium chloride, white precipitate of silver chloride and aqueous calcium nitrate are formed.

   (i) Write a net ionic equation for the reaction

   (ii) State with reason, the type of reaction which took place

   (b)(i) Oxygen gas may be produced in the laboratory by heating a mixture of potassium chlorate with manganese IV oxide. Which of these two compounds produces the required oxygen? Give reason to support your answer.

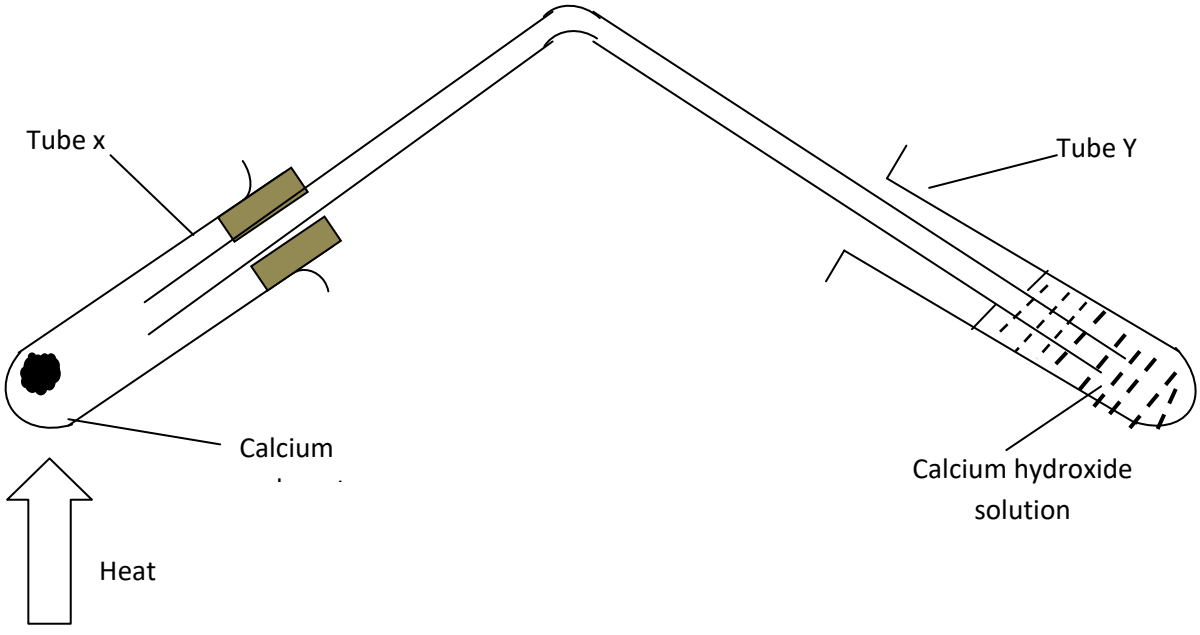
   (ii)Briefly explain why this method is not much used?

   (iii)Give a balanced equation for this reaction in (b)(i) above

5. (a) Elements are classified into metals and non – metals. Basing on the characteristics of metals, relate the characteristics of metals to its uses by giving four uses

   (b) With aid of diagram differentiate three states of matter

6. (a) A student set up the apparatus shown below to investigate the action of heat on a sample of calcium carbonate.



(i) State the observations made in test tubes X and Y

(ii) Write an equation to show the reactions in tube Y after letting the experiment proceed.

(b) During an experiment on the reduction of an oxide of copper, the following data was obtained
Mass of empty boat =25.00g
Mass of empty boat + oxide of copper = 29.00g
Mass of boat + copper (after reaction) = 28.20g
Work out for empirical formula, given that (Cu = 64 and O =16)

7. (a) James was assigned to clean chemistry laboratory by his chemistry teacher. Accidentally he was cut on his leg, his friend assisted him by applying antiseptic solution.
   (i) What is the name given to service given by James's friend?
   (ii) Give out four the importance of the service named in (a) (i) above
   (b)Give the scientific fact about the following statements.
   (i) Water is added to hydrate copper II sulphate salt, no further colour change was observed.
   (ii) Hydrogen gas is used in the production of prestige and margarine
   (iii) Water is universal solvent
   (iv) Hydrogen is found higher in the atmosphere

8. (a) A student was preparing food for the family using hot oils on a frying pan. Accidentally, the pan tipped over and a huge fire spread on the kitchen floor.
   (i) Which fire extinguishers would be suitable for putting out the fire? Explain
   (ii) Which fire extinguishers would not be suitable for putting out the fire? Explain
   (b)Changes of matter can be used in many fields as it gives us useful products. Explain by giving four importance of changes of matter.

SECTION C (30 Marks)
Attempt two questions from this section

9. (a). draw a well labeled diagrams of chemistry laboratory apparatus which are used to measure fixed volume of liquid. (Two apparatus).
   (b). explain five application of volumetric analysis.
10. (a).Define the following terms
    i. Mole
    ii. Molar mass
    (b) 112dm3 of oxygen was collected at STP when a sample of lead nitrate was completely by heat. Calculate the volume of nitrogen dioxide produced

    (c). the equation below shows the dissociation of sulphuric acid into ions.
    $H_2SO_{4\,(aq)} \longrightarrow 2H^+_{(aq)} + SO_4^{2-}{}_{(aq)}$
    From, the equation, how many ions are there in 9.8g of sulphuric acid?

11. Catherine is a student at Kilosa secondary. She was suffering from heartburn and a doctor prescribed to her to use antacids tablets to reveal the condition. By giving six points, explain the applications of the reaction which took place in Catherine's stomach after being given with antacids tablets.